Winning Back The Plate: Predicting And Preventing Churn At PlatefulNZ

Introduction

The award-winning chef Sage Cooke founded PlatefulNZ, a subscription meal kit company serving all of New Zealand. The company faces rising customer attrition and uncertainty about churn drivers. This report analyses the dataset to identify churn predictors and provide actionable recommendations for improving retention and business growth.

Data Overview

The dataset contains 200,000 customer records and 26 variables covering demographics, engagement, subscription details, and retention outcomes.A user_ID uniquely identifies each customer, and retained_binary is the primary target variable (1 = retained, 0 = churned). This dataset is highly relevant for churn prediction because, on average, 76.8% of customers were retained while 23.2% churned from Table 1. While subscription details (plan type, weekly meals, and size) capture lifestyle preferences,(from figure 1 and 2)we can see that demographic features like age (ranging from 17 to 91, averaging 42 years), gender, and location provide insights into customer segments. Email open rates (74%), app and website visits, and social media activity with low overall interaction measure behavioural engagement. Purchase behaviour and loyalty patterns are reflected in transactional details, such as the number of purchases, weeks since the last purchase, add-on purchase value, and discount usage. Additional customer experience indicators include support interactions, payment problems (5.4%), and satisfaction survey results. Due to its extensive coverage of behavioural, demographic, and transactional factors, this dataset offers a solid basis for modelling customer attrition and creating practical retention plans for PlatefulNZ.We started cleaning the dataset with base were week since last sign up.

To better capture consumer behaviour, we developed new features in addition to raw variables. Likes, shares, posts, and comments were combined to create a weighted social media score representing each customer's online engagement degree. Building on this, we created an engagement band that blends the social media score with other important activity metrics, such as the quantity of purchases, app and website visits, and email responsiveness we can see that in figure 3. Using this composite metric, we could categorise customers into various engagement levels, which made it simpler to spot trends in loyalty and churn risk. These changes improved the dataset's capacity for prediction and offered a more comprehensive perspective on consumer interaction.

Table 1: Proportion of Customers by Retention Status

Var1	Freq
0	0.232
1	0.768

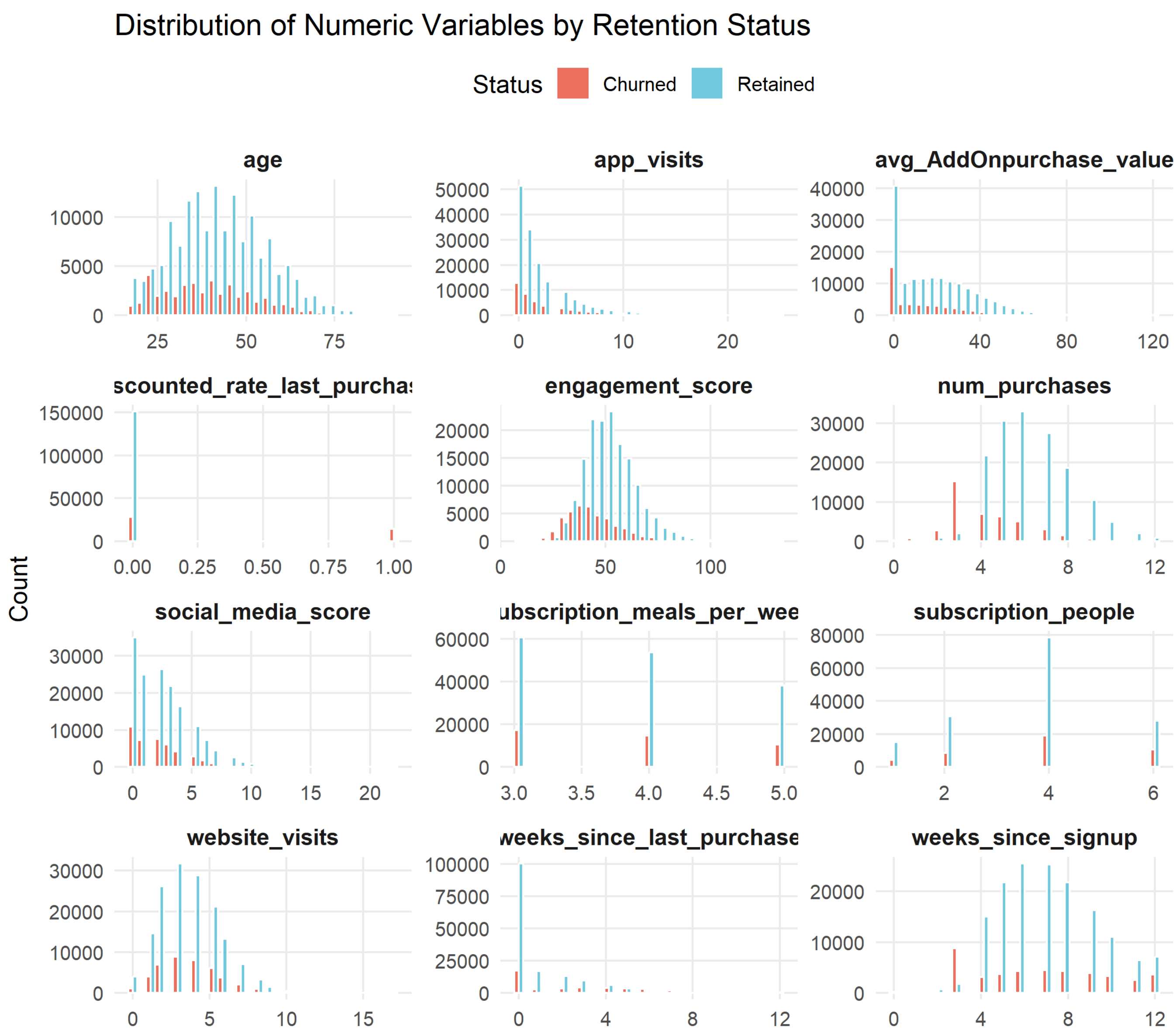
Assumptions:

We made several assumptions to guide the analysis:

• Churn was defined strictly as subscription cancellation, not temporary pauses.

• Social media interactions were considered equally weighted in constructing the engagement score.

Figure 1 :Numeric Variables Distribution based on retention status



Exploratory Data Analysis (EDA) and Statistical Testing :

EDA revealed several potential churn predictors. Younger customers appeared more likely to churn, while older age brackets showed stronger loyalty. Satisfaction scores were clearly associated with retention, supporting the idea that service quality is critical. Formal statistical tests confirmed many of these relationships. Table 2 summarises results across multiple variables. Engagement, satisfaction, and payment issues were all highly significant predictors ($p < 0.001$). In contrast, gender showed no statistical relationship with churn. Table 2. Results of statistical tests on predictors of churn. A correlation analysis among numerical predictors (Figure 4) further highlighted meaningful relationships. Satisfaction was positively correlated with retention, while higher payment issues correlated negatively with retention. Importantly, multicollinearity was not a major concern, suggesting predictors could be used together in regression models. Correlation among numerical predictors. The ggpairs plot (Figure 5) reinforced these findings by showing pairwise patterns. In particular, engagement and satisfaction consistently separated churned customers from retained ones. Figure 5. Pairwise relationships among selected predictors.

Figure 2:Categorical variable Distriburion based on retention status

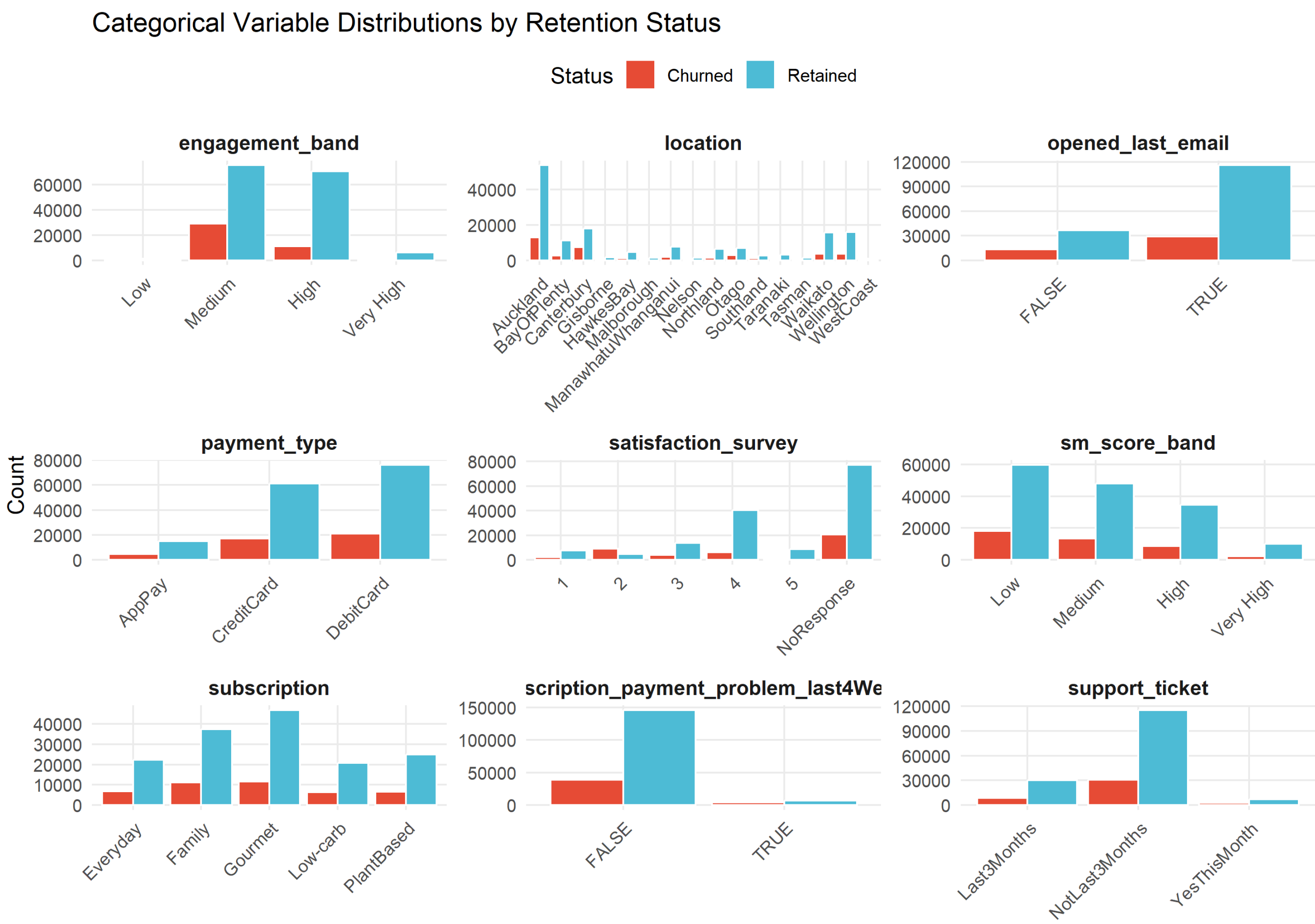
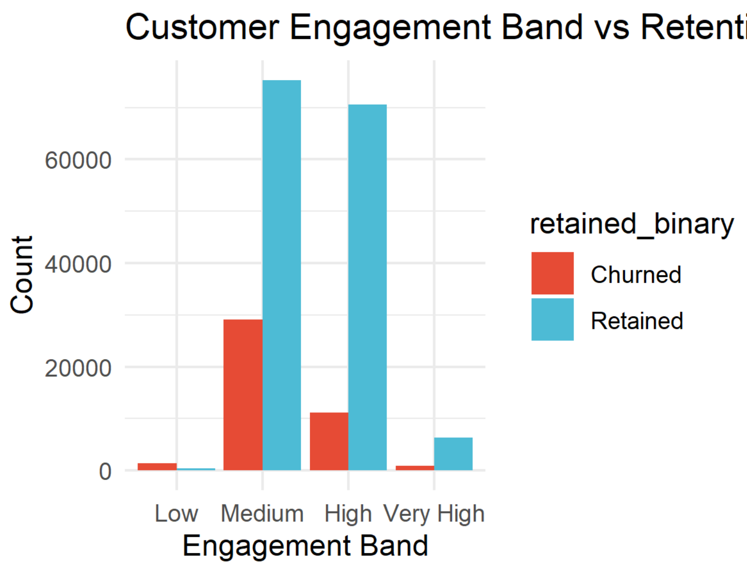


Figure 3: Shows Engagement band based on retention status



Methodology

We applied three classification techniques to model churn:

1. Logistic Regression (GLM) – chosen for its interpretability and ability to highlight significant predictors.
2. Neural Network (MLP) – included to capture potential non-linear relationships.
3. LightGBM (Gradient Boosting) – selected as a modern, high-performing algorithm suitable for imbalanced datasets.

The dataset was split 80/20 into training and test sets. Categorical predictors were dummy-encoded, numerical variables were normalised, and class imbalance was mitigated via up-sampling. Model performance was assessed using confusion matrices, accuracy, sensitivity, specificity, and AUC (Area Under the ROC Curve).

Table 2: Statistical Test Results

variable	type	t_test_p	levene_p	chisq_p
age	numeric	0.0000	0.0000	NA
support_ticket	categorical	NA	NA	0.0000
satisfaction_survey	categorical	NA	NA	0.0000
location	categorical	NA	NA	0.0000
subscription	categorical	NA	NA	0.0000
subscription_meals_per_week	numeric	0.0636	0.0649	NA
subscription_people	numeric	0.0000	0.0000	NA
payment_type	categorical	NA	NA	0.0017
app_visits	numeric	0.0000	0.0000	NA
website_visits	numeric	0.0000	0.1349	NA
opened_last_email	categorical	NA	NA	0.0000
subscription_payment_problem_last4Weeks	categorical	NA	NA	0.0000
weeks_since_signup	numeric	0.0000	0.0000	NA
num_purchases	numeric	0.0000	0.9717	NA
weeks_since_last_purchase	numeric	0.0000	0.0000	NA
discounted_rate_last_purchase	numeric	0.0000	0.0000	NA
avg_AddOnpurchase_value	numeric	0.0000	0.0000	NA
social_media_score	numeric	0.0000	0.0000	NA
sm_score_band	categorical	NA	NA	0.0000
engagement_score	numeric	0.0000	0.0000	NA
engagement_band	categorical	NA	NA	0.0000

Figure 4: Correlation analysis plot

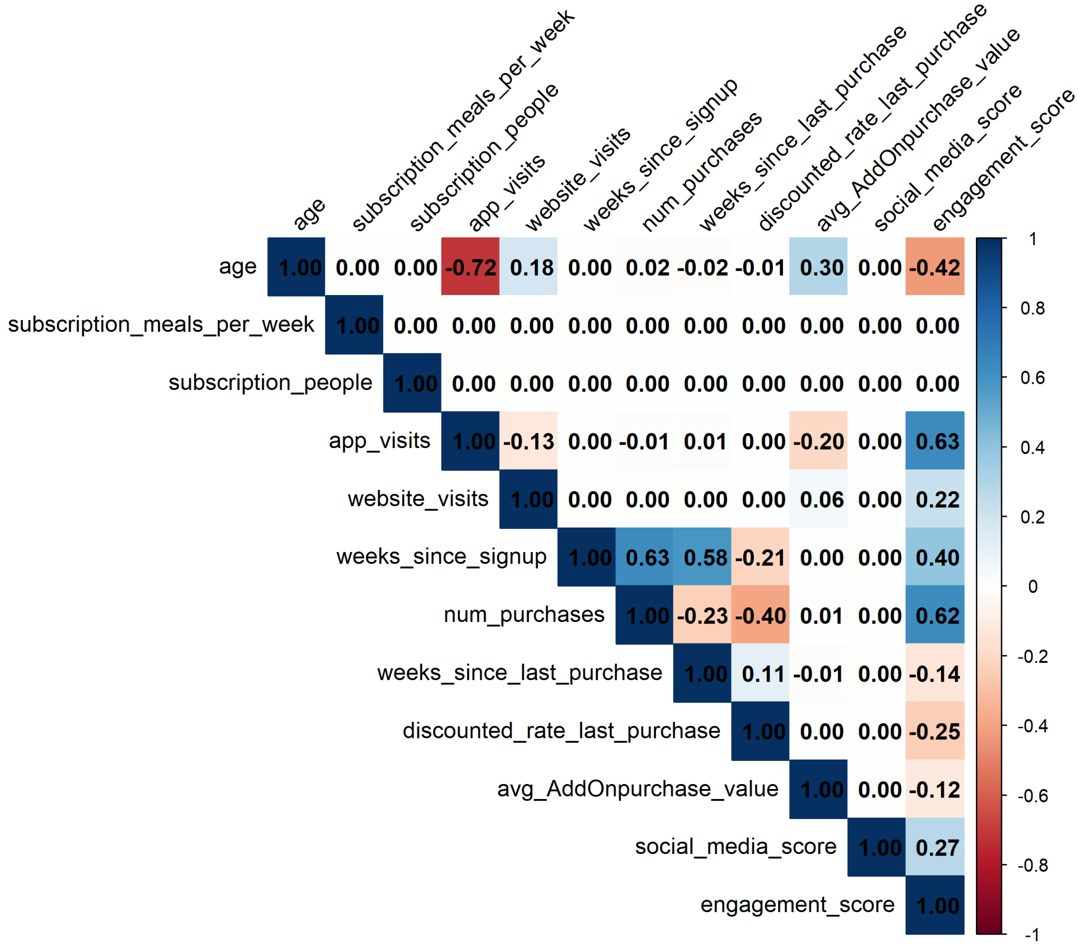


Figure 5:Pairwise relationship of key variable based on retentation status

Results

Table 3 present the performance metrics of each model.

• Logistic Regression achieved solid interpretability, with sensitivity of 0.81, specificity of 0.93, and an AUC of 0.92(from figure 6).

• The Neural Network slightly improved sensitivity (0.84) but at the cost of lower specificity (0.88), with an AUC of 0.91(from figure 7).

• LightGBM outperformed both, with sensitivity of 0.87, specificity of 0.93, and the highest AUC at 0.953(from figure 8).

The results showed consistent patterns across models, with engagement, payment issues, and satisfaction being the most influential predictors.Out of all the models LightGBM showed us the best results.The table 4 shows the confusion matrix of the best model which is LightGBM.

The LightGBM model performed best overall,(from figure 9) balancing accuracy with sensitivity which is 72.5%(ability to correctly identify churners).High specificity across all models indicates strong ability to detect retained customers, but improving sensitivity was crucial for churn prediction.The results confirmed that engagement behaviours, billing experiences, and satisfaction scores were the strongest predictors of churn, aligning with the statistical tests.The high AUC (0.953) confirms that LightGBM is robust across thresholds, making it the best model for operational churn prediction.

Insights

Based on the statistical analysis and model outputs, three primary insights emerged:

1.Engagement is the strongest retention driver Customers who log in frequently, browse recipes, or purchase add-ons show significantly higher retention. The engagement band (Figure 3) and test results (Table 2) both confirmed that disengaged customers are far more likely to churn.

2.Satisfaction drives loyalty Low satisfaction strongly predicted churn across all models. Unresolved complaints or poor experiences reduce trust and increase attrition risk, making satisfaction one of the most actionable levers for retention.

3.Operational frictions accelerate churn Payment failures and billing issues disproportionately increased churn, even among otherwise engaged customers. Streamlining payment processes and quickly resolving service issues are essential to customer retention.

4.Demographics play a secondary role While some demographic groups showed slightly higher churn, their impact was minor compared to engagement and satisfaction. This confirms that churn at PlatefulNZ is mainly behavioural and operational.

LightGBM is the most reliable predictive model With an AUC above 0.95, LightGBM outperformed other models, providing accurate and scalable predictions for operational deployment.

Figure 6:Shows logistic regression roc curve

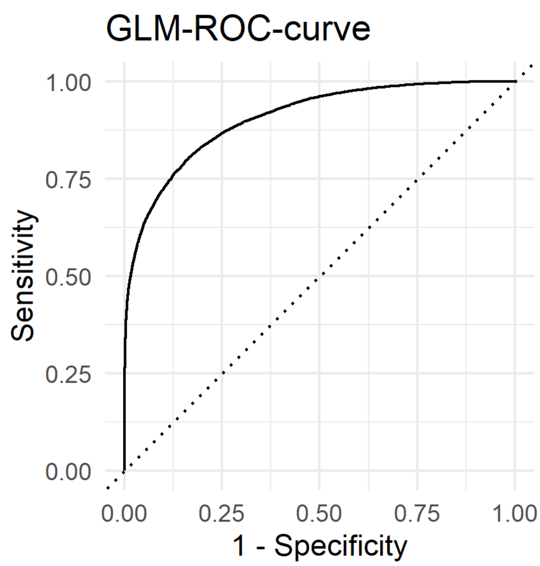


Figure 7:Shoes LightGBM ROC curve

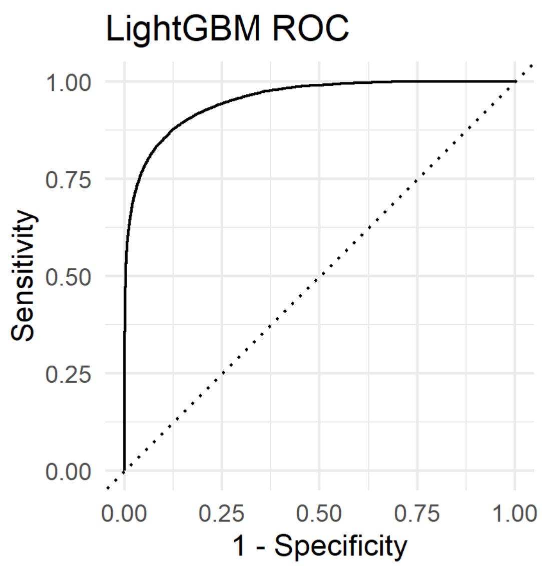


Figure 8:Shows the neural net AUC curve

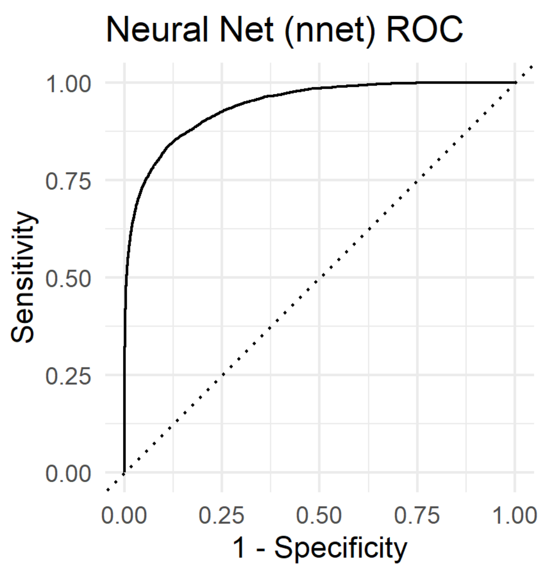


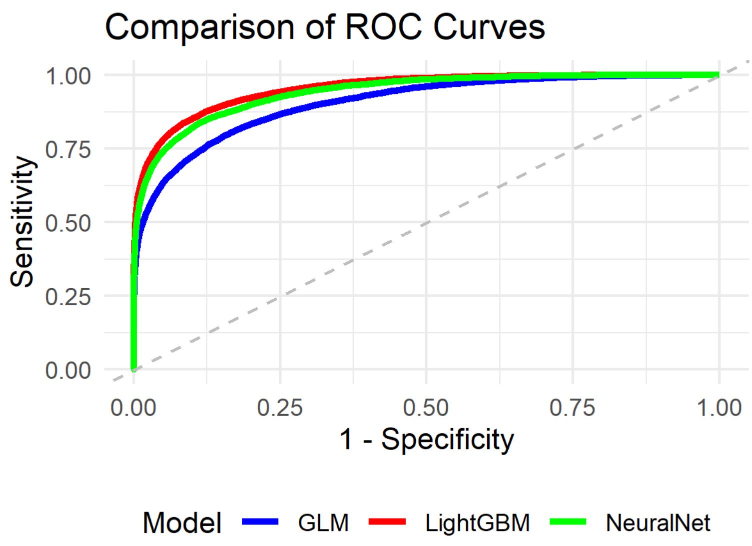
Table 3: Model Performance Comparison of metrics

Model	Accuracy	Sensitivity	Specificity	AUC
Logistic Regression	0.883	0.595	0.963	0.906
Neural Net	0.908	0.698	0.966	0.942
LightGBM	0.917	0.725	0.970	0.953

Table 4: Confusion Matrix: Prediction vs Truth

Prediction	Churned	Retained
Churned	6174	919
Retained	2339	29595

Figure 9: shows the comparision of models AUC curve



Recommendations

The following recommendations are directly grounded in the insights from the analysis. Each is designed to be actionable, measurable, and strategically aligned with PlatefulNZ's business goals.

1. Boost Digital Engagement :Introduce personalised recipe recommendations, gamified cooking challenges, and loyalty rewards for frequent app users. Our models show engagement is the most significant predictor of retention. By fostering habitual interaction, PlatefulNZ can increase switching costs and keep customers invested in the service.

2. Implement Proactive Payment Support :Introduce automated payment reminders, retry systems, and flexible payment options. Since Customers experiencing payment failures are at high risk of churn. Addressing billing issues proactively will remove a key operational friction and directly reduce avoidable churn.

3. Launch a Satisfaction Recovery Program :Monitor satisfaction scores in real time and trigger retention campaigns (discounts, free add-ons, or personalised outreach) when scores drop. Because Dissatisfied customers were significantly more likely to churn. Targeted recovery strategies can re-engage at-risk customers before cancellation occurs.

4. Encourage Add-On Purchases: Promote seasonal add-ons and bundle offers to increase customer spend. Since Add-on purchases are associated with higher retention, likely because they increase perceived value and deepen commitment to the service. Each recommendation is evidence-based: it stems directly from the variables identified as churn drivers in both statistical testing and predictive modelling.

Conclusion

Our analysis shows that PlatefulNZ's churn is driven mainly by low engagement, dissatisfaction, and payment issues, not demographics. LightGBM proved the best model with over 91% accuracy, high AUC, and strong sensitivity—ensuring most at-risk customers are identified. Focusing on engagement, satisfaction recovery, and payment reliability will help PlatefulNZ cut churn, boost lifetime value, and secure sustainable growth. If implemented effectively, the recommended strategies have the potential to significantly reduce attrition, improve customer lifetime value, and strengthen the company's position in the competitive meal kit market.